Excitation and fragmentation of C_nN^+ (n=1-3) molecules in collisions with He atoms at intermediate velocity; fundemental aspects and application to astrochemistry

Soutenance de thèse

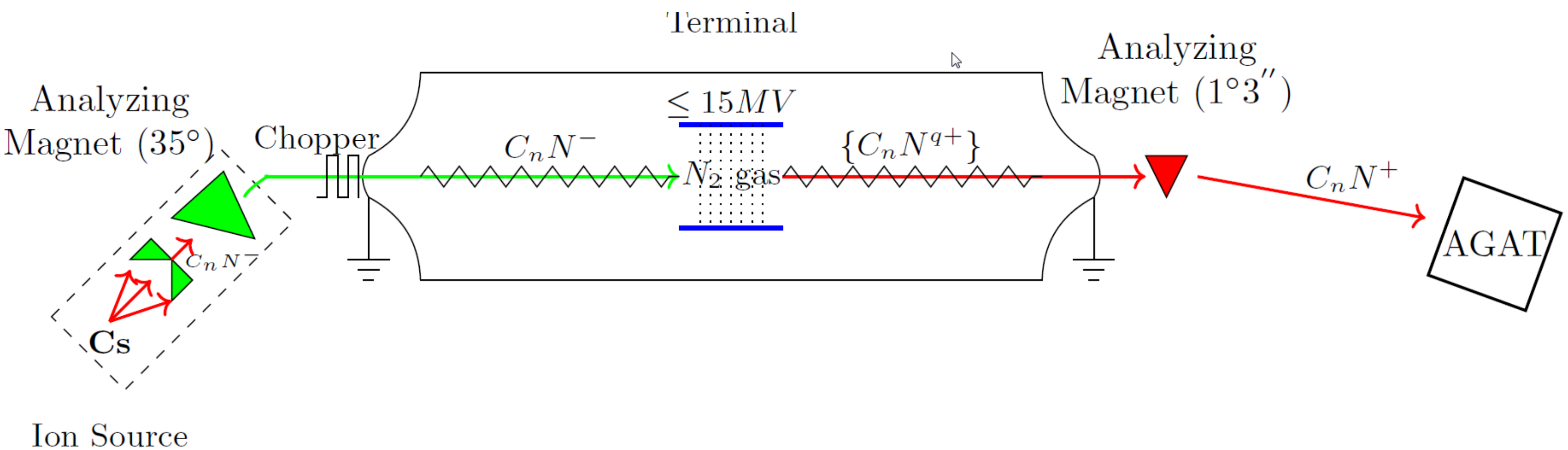
Sous la direction de Karine Béroff

Le 28 septembre 2018

Thejus Mahajan
ISMO

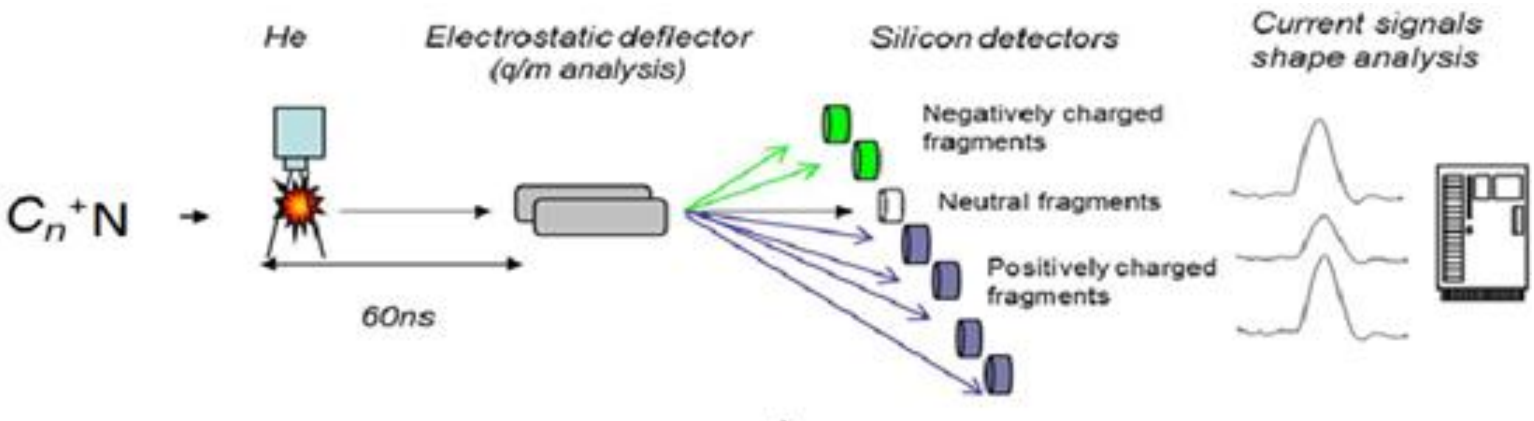
- Plan
- Introduction
- **Experimental tools**
- Modelisation of the collision
- Comparison between experiment and modeling for cross sections
- Application of dissociation branching ratios measurements to astrochemistry
- Conclusions and perspectives

General view of experimental set-up



- Sputtering source for C_nN^- - Mixture of TiN and graphite sputtered with Cs^+.
- Magnetic selection and initial acceleration up to terminal
- Stripping through N_2 gas.
- C_nN^+ accelerated up to v_p= 2.2 a.u till AGAT.

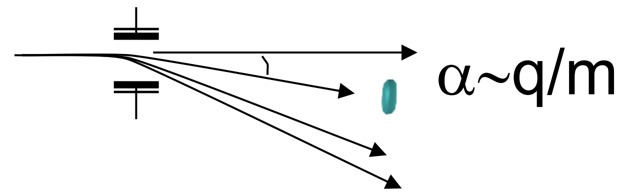
The AGAT setup



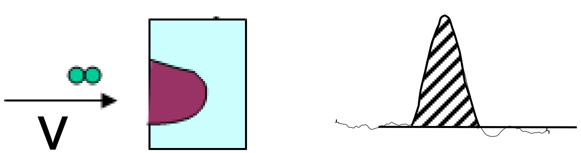
- Collision on a gaseous He jet
- Analysis of projectile after collision with electrostatic analyser according to q/m ratio
- Detection of fragments positive/negative/neutrals
- Dedicated software NARVAL.

Identification of the projectile charge after collision

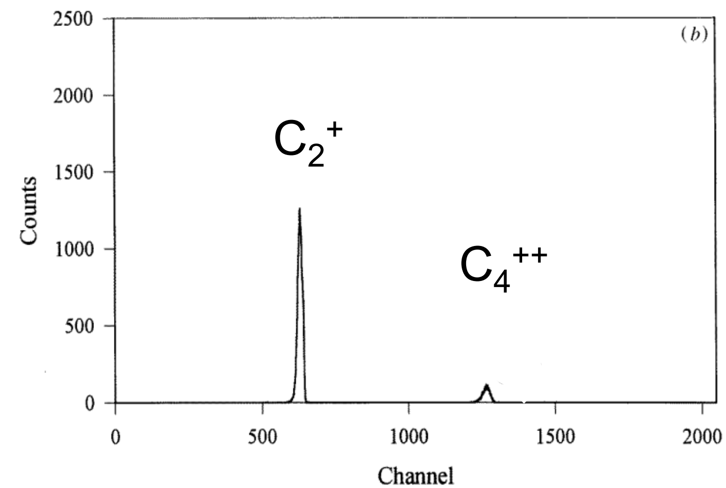
Dectectors are positioned as q/m ratio



Energy deposited on each detector : m

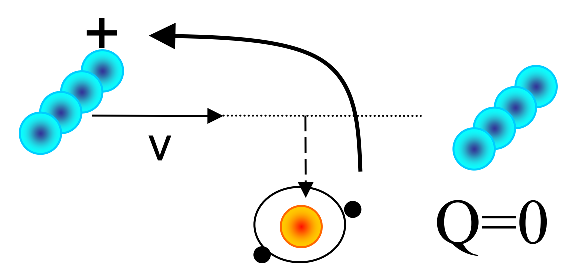


$E = mv^2/2$
(v=cte)



Coincidences between all the detectors: $Q_{TOT} = \sum_f q_f$

Q=0 Single electron capture
Q=-1 Double electron capture
Q=1 Electronic excitation
Q=2 Single ionization
Q=3 Double ionization….



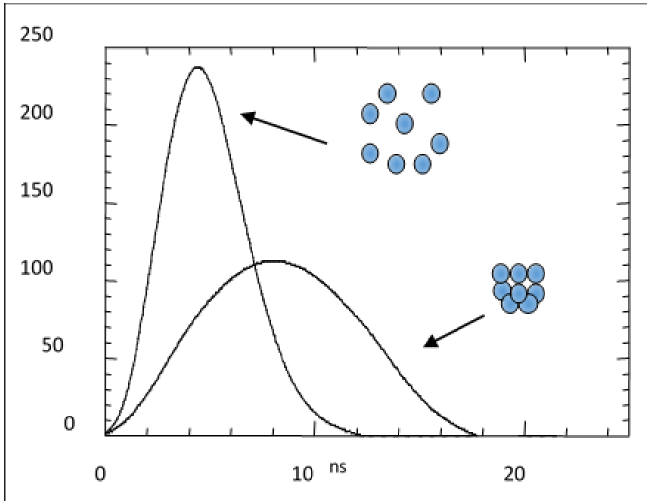
Resolution of the fragmentation

- Coincidences between fragments impinging onto different detectors : $X^p/Y^q/Z^r$

- Fragments impinging in the same detector (Neutrals):

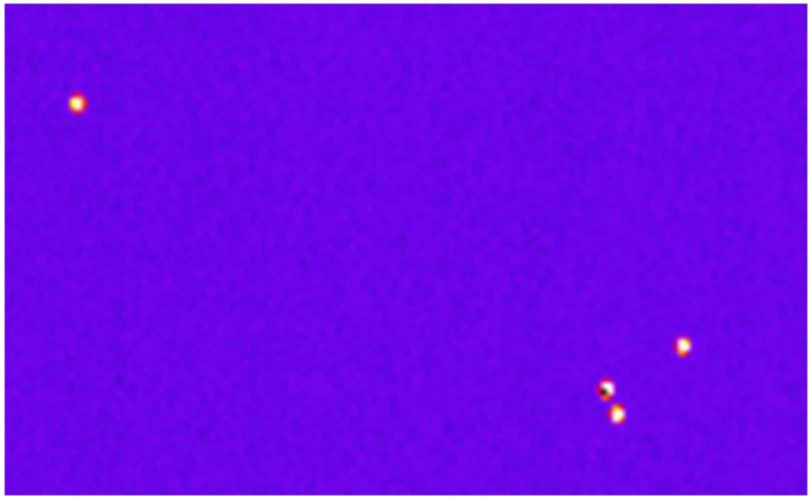
Shape analysis method

(Chabot et al 2002)



CCD camera (chabot et al 2011)

Made of 512x512 pixels 24μm each; combines the mass recognition and position informations



Visualization of C2N + C dissociation channel

Dissociation branching ratios

- For a process, molecule after collision can fragment or not.
- Each product are different branches of same process.
- Fractional contribution of each branch is its branching ratio.

- Example single capture : $C_2N^+ \; + \; He \rightarrow \{C_2N\}$

  $\{C_2N\}$ = C/C/N, C_2/N, C/CN, C_2N (intact)

- Branching ratio for a given channel is

$$BR_{channel} = \frac{P_{channel}}{\{P_{C_2N}\}}$$

$P_{channel}$ is the probability of 'channel' and $\{P_{C2N}\}$ is the total probability of single capture

Determination of absolute beam-jet overlap

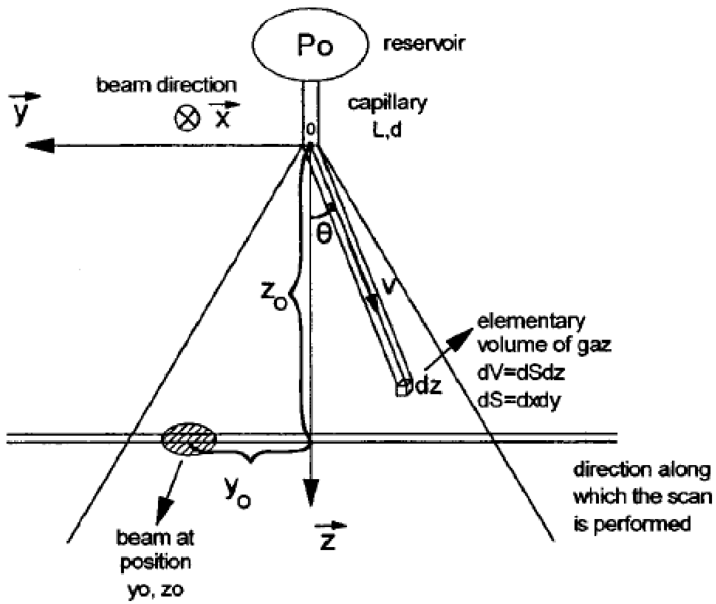


Fig1: View of jet and beam relative positions

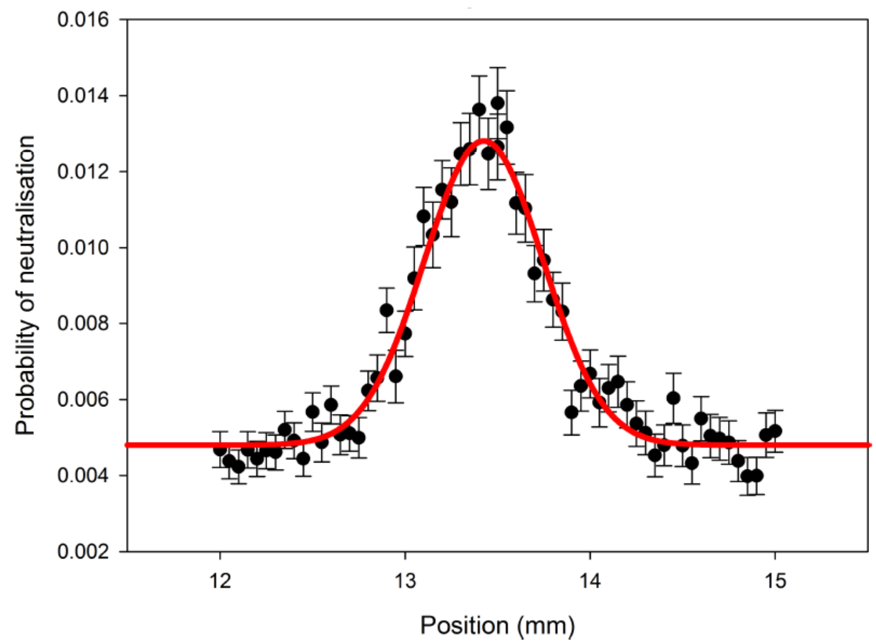


Fig2 : Example of jet profile

$$\sigma = \frac{prob(y_0, z_0)}{B_{jet}(y_0, z_0)}$$

$$\frac{prob(y_0, z_0)}{\int prob(y_0, z_0)dy_0} = \frac{B_{jet}(y_0, z_0)}{\int B_{jet}(y_0, z_0)dy_0}$$

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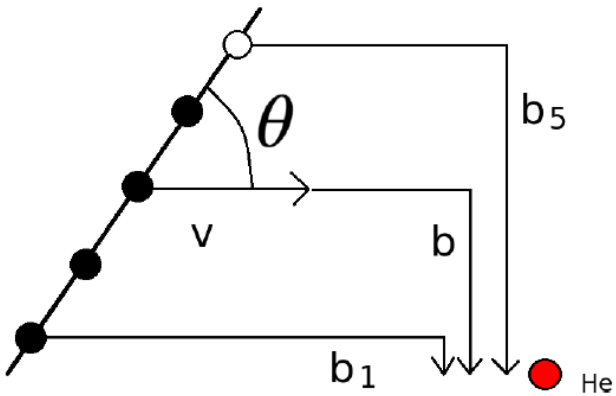
The Independant Atom and Electron (IAE) model



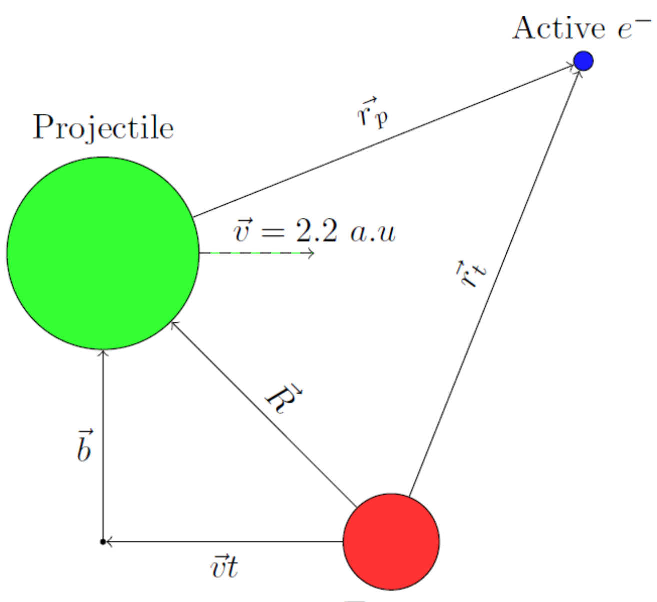
Collision is treated as five independent collisions of the atom-atom or ion-atom type, each one at its own impact parameter

Independent atoms: $$P_{Neutr}(\boldsymbol{b}) = \sum_{i}^{n} P_{capt}^{(1)}(b_i) \times \prod_{i=1}^{n}(1 - P_{ion}(b_i)) \prod_{j\neq i}(1 - P_{capt}(b_j))$$

Independent electrons:
$$P_{capt}^{(1)}(b_i) = 2p_c(b_i)\big(1 - p_c(b_i)\big)$$
$$1 - P_{capt}(b_i) = \big(1 - p_c(b_i)\big)^2$$
$$\big(1 - P_{ion}(b_i)\big) = \left(1 - p_{ion}^{(2s)}(b_i)^2\right)\left(1 - p_{ion}^{(2p)}(b_i)\right)^2$$

Cross section: $\sigma_{proc} = \int\int\int P_{proc}(b,\theta,\alpha) bdbsin\theta d\theta d\alpha$ (Study and reflect if sin and dθ is necessary)

The Classical Trajectory Monte Carlo (CTMC) method for the atom(ion)-atom collision



- Impact parameter formalism
- Classical distribution of electron density

$$\rho(\boldsymbol{r}, \boldsymbol{p}, t)$$

- Satisfies the Liouville equation (*)

$$\frac{\partial \rho}{\partial t} = -[\rho, H_e] = -\frac{\partial \rho}{\partial \boldsymbol{r}}\frac{\partial H_e}{\partial \boldsymbol{p}} + \frac{\partial \rho}{\partial \boldsymbol{p}}\frac{\partial H_e}{\partial \boldsymbol{r}}$$

H_e is the electronic Hamiltonian

* Classical equivalence of Schrödinger equation

Electronic hamiltonians for 1e$^-$ and 2e$^-$ calculations

- 1e$^-$ calculations: active e$^-$ on target (target ionization, electron capture)

$$H_{1e^-} = \frac{p^2}{2} + V_{mod}^{He^+ + e}(r_t) + V_{mod}^{C+e}(r_p) \qquad \text{(C – He collision)}$$

- 1e$^-$ calculation: active e$^-$ on projectile (projectile excitation and ionization)

$$H_{1e} = \frac{p^2}{2} + V_{mod}^{He+e}(r_p) + V_{mod}^{N^+ + e}(r_t) \qquad \text{(N – He collision)}$$

- 2e$^-$ calculations : 1e$^-$ on target and 1e$^-$ on the projectile

(projectile ionization and excitation)

$$\begin{aligned} H_{2e^-} = {} & \frac{p_1^2}{2} + V_{mod}^{(He^+ + e)}(r_{1T}) + V_{mod}^{(X^{2+} + e)}(r_{1P}) \\ & + \frac{p_2^2}{2} + V_{mod}^{(He^+ + e)}(r_{2T}) + V_{mod}^{(X^{2+} + e)}(r_{2P}) \\ & + \frac{1}{|\boldsymbol{r}_{1T} - \boldsymbol{r}_{2T}|} \end{aligned}$$

(X^+ + He collision)

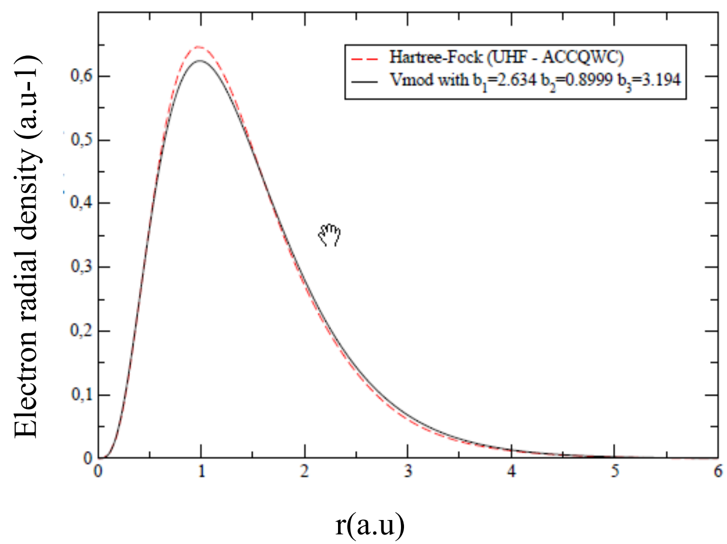
The model potentials

- ✓ For e^- - *He* system : $V_{mod}(He^+ + e^-) = -\dfrac{Z-N}{r} - \dfrac{1+\alpha r}{r} e^{-2\alpha r}$ (α=1.6875)

- ✓ For C, C^+, N, N^+ - *He* system : $V_{mod}(r) = -\dfrac{Z-N}{r} - \dfrac{Ae^{-Br} + (N-A)e^{(-Cr)}}{r}$

Orbital	$NIST$	V_{mod}
2p	-0.5345	-0.5346
3s	-0.1548	-0.1541
3p	-0.1004	-0.1009
3d	-0.0568	-0.0581
4s	-0.0621	-0.6240
4p	-0.0472	-0.0474
4d	-0.0319	-0.0326

A, B and C chosen so as to matchwith the NIST values.



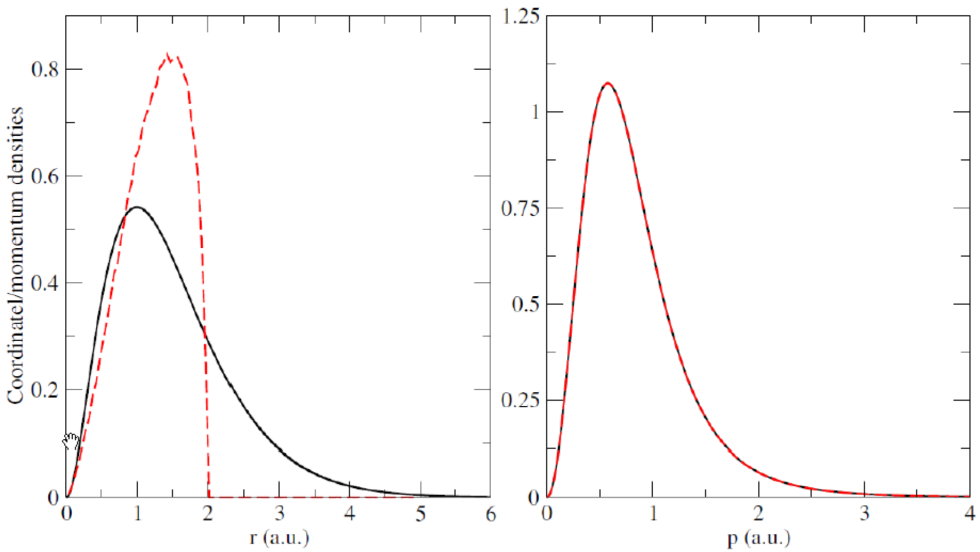
Comparison between 2p radial densities With V_{mod} and with Hartree-Fock

Projectile	Z	N	A	B	C
C	6	6	1.964	7.136	0.840
C^+	6	5	1.904	0.808	2.518
N^+	7	6	2.634	0.899	3.194
N^{++}	7	5	2.401	1.264	3.238
C^{++}	6	4	2.044	1.256	3.202

Values of A,B,C used in this work

Initial electronic distribution functions

$$\rho(\boldsymbol{r},\boldsymbol{p},t \to -\infty) = \sum_{i=1}^{N} \delta(E_0 - E_i(t \to -\infty))$$



- N non-interacting electrons having all the same energy E_0 : Microcanonical distribution
- Not good for the classical radial density
- Limitations may arise when dynamical processes involve electron located at the tail of the quantum mechanical density.

Comparison between **quantum** (black) and **microcanonical** (red) distributions of H(1s)

Two- electron calculations

$$\rho(\boldsymbol{r_1},\boldsymbol{r_2}.\boldsymbol{p_1},\boldsymbol{p_2},t \to -\infty) = \rho_{proj}(\boldsymbol{r_1},\boldsymbol{p_1})\rho_{tar}(\boldsymbol{r_1},\boldsymbol{p_2})$$

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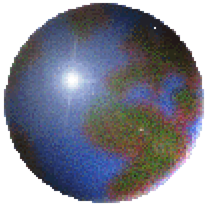

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Molecules in the interstellar medium

- First discovery of CH in ISM (1937)
- Today, more than 200 molecules detected, mostly neutrals but also cation and anions.
- C_nN (n=1,3,5) and C_nN^- (n=1,3,5) have been detected in ISM and also in planetary atmospheres

Molecules are especially abundant, together with dust, in dense molecular clouds such as the one shown here

Radiations and induced dissociation processes

- Molecules are submitted to various radiations (photons, cosmic-rays (p, He^+…), electrons) that induce dissociation of the molecules.

- Main dissociative processes:
  - Photodissociation: $AB + h\nu \rightarrow A + B$
  - Dissociation induced by cosmic rays: $AB + CR \rightarrow A + B$
  - Dissociative recombination: $AB^+ + e^- \rightarrow A + B$
  - ❖ Ion-molecule reaction: $A + B^+ \rightarrow C^+ + D$
  - ❖ Neutral-neutral reaction: $A + B \rightarrow C + D$
  - ❖ Charge exchange with He^+: $AB + He^+ \rightarrow A^+ + B + He$

  - ❖ These last three are bimolecular processes involving the abundances of two molecules (or atoms) in their probability to occur.

Kinetic databases for astrochemistry

The abundance of molecules are governed by kinetic equation such as:

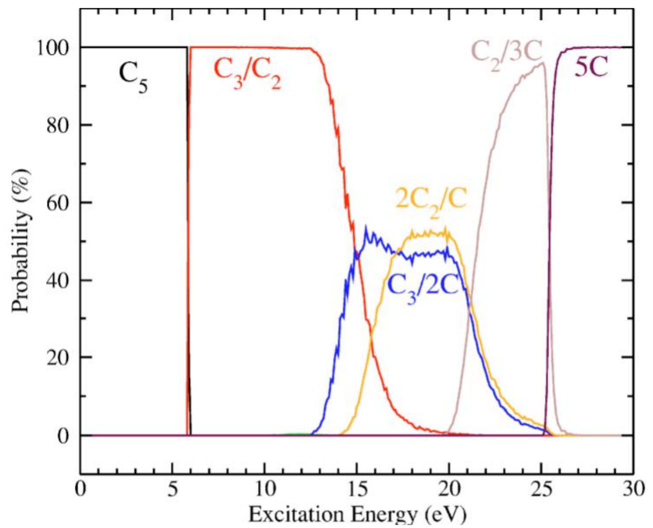
$$\frac{dn_i}{dt} = \sum_{j,k} k_{jki} n_j n_k + \sum_{l} \zeta_{li} n_l - \sum_{l} \zeta_{il} n_i - \sum_{j,l} k_{ijl} n_i n_j$$

- n_i is the density of species i, populated by the two first terms of rhs of equation and depopulated by the two last ones.
- Bimolecular and unimolecular process contribute with rate coefficients k_{xyz} and ζ_{xy}.
- Kinetic databases such as KIDA (http://kida.obs.u-bordeaux1.fr) are hosting values of rate coefficients for many reactions and species. Branching ratios for different products of the same reactants are badly lacking.

- We intend to provide dissociation BR to the kinetic databases.

The concept of Breakdown curve

- In the frame of statistical fragmentation, dissociation only depends on the internal energy E.
- The curves providing the dissociation branching ratios BR_j as a function of E are called breakdown curves $BDC_j(E)$



Diaz-Tendero et. al (2005)

- BDC are specific to each molecule
- BDC allow to predict dissociation BR if $f(E)$ is known:

$$BR_j = \int_0^{\infty} BDC_j(E) f(E) dE$$

Construction of BDC by a semi-empirical method

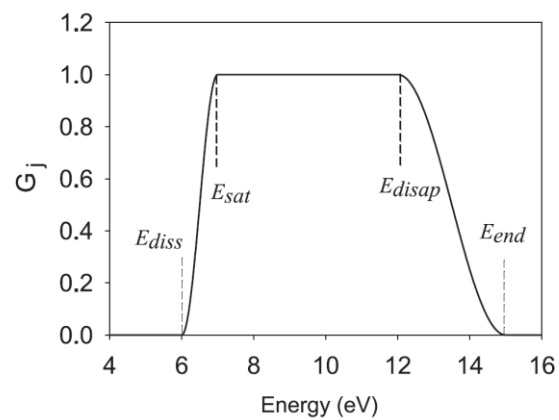
$$BR_j = \int BDC_j(E) f(E) dE \quad (1)$$

- BR_j is measured, $f(E)$ is known : Inversion of (1) → $BDC_j(E)$

- $BDC_j(E)$ is written as:

$$BDC_j(E) = \frac{a_j G_j(E)}{\sum_j a_j G_j(E)} \qquad (2)$$

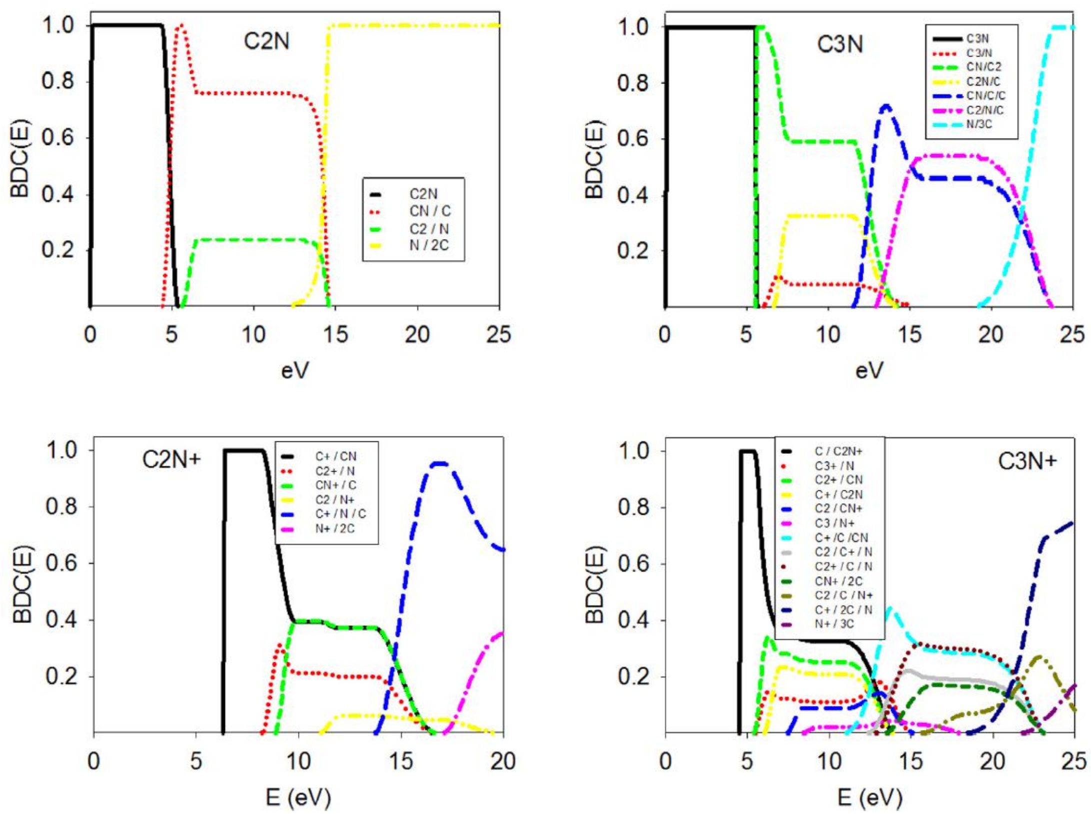
$$\sum_j BDC_j(E) = 1 \qquad (3)$$



E_{diss}, E_{sat}, E_{disap}, E_{end} are obtained from the theory (Diaz-Tendero et al)

We have to solve j equations of type (1) that are coupled through (3).
a_j values are obtained by a minimization procedure.

Semi-empirical BDC's for C_2N, C_3N, C_2N^+ and C_3N^+



The predictions are more robust in flat domains as compared to curve crossings.
For cations, we begin at E_{diss} because the intact rate is not measured.

Predictions of products BR of astrochemical interest

- We can predict dissociation BR of $C_nN^{(+)}$ $(n = 2 - 3)$ adducts
- Example of ion-molecule reactions given in the Table below

$E_a = E_{diss}$ (adduct → reactants) ; $\Delta E = E_a - E_{diss}$ (adduct → products)

Reactants	Products	Model BR (abs.err)	KIDA BR	E_a(eV)	ΔE (eV)
$\mathbf{C_2^+ + N}$	$C^+ + CN$	1	1	**8.24**	1.85
$\mathbf{CN^+ + C}$	$C^+ + CN^*$	$0.72(+0.07/-0.05)$	1	**8.86**	2.47
	$C_2^+ + N^*$	$0.28(+0.05/-0.07)$			0.62
$\mathbf{N^+ + C_2}$	$CN^+ + C^*$	0.40(0.05)		**11.02**	2.16
	$C^+ + CN^*$	0.39(0.05)			4.63
	$C_2^+ + N^*$	0.21(0.04)	1		2.78
$\mathbf{C_3^+ + N}$	$[C_2N^+] + C$	1		**5.45**	0.93
$\mathbf{C_2^+ + CN}$	$[C_2N^+] + C$	1		**5.43**	0.91
$\mathbf{C^+ + CNC}$	$[C_2N^+] + C$	0.58(0.06)		**6.02**	1.5
	$C_2^+ + CN$	0.30(0.06)			0.59
	$C_3^+ + N$	0.12(0.04)			0.57
$\mathbf{C^+ + CCN}$	$[C_2N^+] + C$	0.58(0.06)	0.7¤	**6.1**	1.58
	$C_2^+ + CN$	0.30(0.06)	0.3		0.67
	$C_3^+ + N$	0.12(0.04)			0.65
$\mathbf{CN^+ + C_2}$	$[C_2N^+] + C$¶	$0.35(+0.03/-0.35)$		**7.4**	2.88
	$C_2^+ + CN$¶	$0.28(+0.03/-0.28)$	1		1.97
	$C^+ + [C_2N]^*$	$0.23(+0.77/-0.03)$			1.38
	$C_3^+ + N$¶	$0.12(+0.02/-0.12)$			1.95
$\mathbf{N^+ + C_3}$	$[C_2N^+] + C$	$0.33(+0.15/-0.04)$		**8.28**	3.76
	$C_2^+ + CN$	$0.25[+0.15/-0.03)$			2.85
	$C^+ + [C_2N]$	$0.22(+0.14/-0.03)$			2.26
	$C_3^+ + N$	$0.11(+0.13/-0.02)$			2.83
	$CN^+ + C_2$¶	$0.09(+0.02/-0.09)$			0.88

Uncertainties on Products BR were obtained by running the code with different inputs within their own error bars. Additional large errors come from spin forbidden reactions that may or may not occur.

Predictions of products BR of astrochemical interest (2)

Example of charge exchange with He+

Ea=IP(He)-IP(molecule) ; ΔE=Ea -Ediss(molecule+products)

Reactants	Products (+ He)	Model BR (abs.err)	KIDA BR	E_a(eV)	ΔE (eV)
CNC + He$^+$	$C^+ + C + N$	0.30(0.15)		**14.93**	1.22
	$C^+ + CN$	0.25(0.10)			8.54
	$CN^+ + C$	0.25(0.07)			6.07
	$C_2^+ + N$	0.15(0.05)			6.69
	$N^+ + C_2$	0.05(0.02)			3.91
CCN + He$^+$	$C^+ + C + N$	0		**13.92**	0.21
	$C^+ + CN$	0.38(0.05)	1		7.53
	$CN^+ + C$	0.38(0.05)			5.06
	$C_2^+ + N$	0.20(0.03)			5.68
	$N^+ + C_2$	0.04(0.02)			2.9
C_3N + He$^+$	$C^+ + C + CN$	0.25(0.04)		**12.81**	1.9
	$C_3^+ + N$	0.18(0.04)			7.36
	$[C_2N^+] + C$	0.15(0.07)			8.29
	$CN^+ + C_2$	0.14(0.03)			5.41
	$C_2^+ + CN$	0.12(0.06)	1		7.38
	$C^+ + [C_2N]$	0.09(0.05)			6.79
	$N^+ + C_3$	0.03(0.01)			4.53
	$C^+ + C_2 + N$	0.02(0.01)			0.56

Many more channels than reported in KIDA

Astrophysical implications

V.Wakelam (LAB Bordeaux)

- The effect of new BR's has been followed in a cold core environment (TMC -1)
- The Nautilus astrochemical code has been used and new BR's introduced.
- No variation above 5% was found for any molecule at any time.

Reasons?

- Number of species and reactions in Nautilus is 700 and more than 10,000; we only changed a few
- Modifications are done on unstrategic reactions. Example:
- CN^+ product is mainly formed by* :

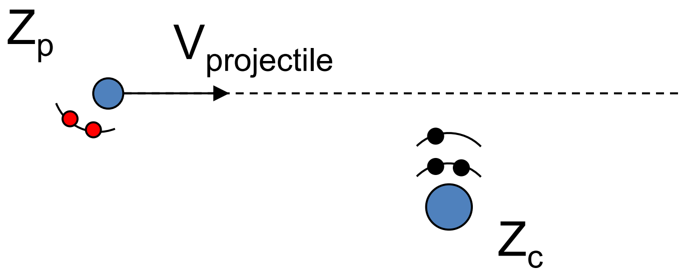
  ionization of CN by CRP (Cosmic Ray Photons)

  $HCN + He^+ \rightarrow CN^+ + H + He$ reaction

  Our reactions for forming CN^+ only represent 3%

*Results obtained at early times ($3\ 10^5$ yrs)

Ion-atom collisions



<u>Velocity regime</u>:

$$K = V_p/V_e$$

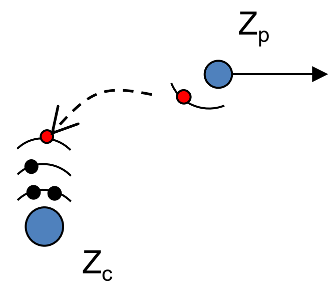
$K << 1$: Elastic diffusion is dominant.

$K \gtrsim 1$: Inelastic processes (electron transitions) dominant

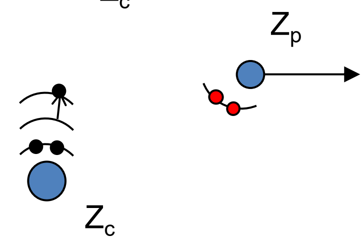
$K \sim 1$: All processes are having the same order of magnitude and are at their maximum of probability.

<u>The electronic processes of importance</u>:

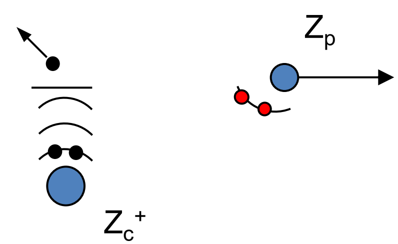
- *Capture* :



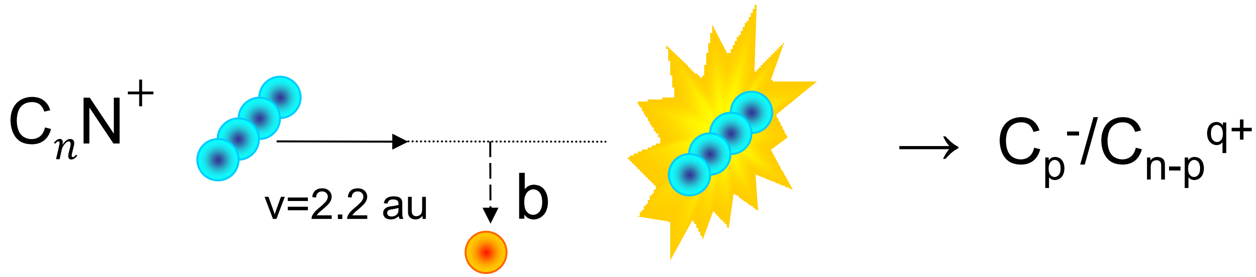
- *Excitation* :



- *Ionisation*:



Molecule-atom collisions



The collision is more complex :

- ❖ The multi-center projectile presents new degrees of freedom (vibration, rotation)
- ❖ More electrons are involved.
- ❖ Once the molecule is excited, it can fragment that complicates the experimental set-up dedicated to the collision study.

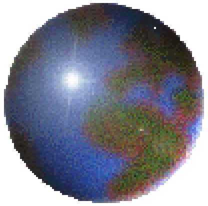
Purpose of this work

- We studied N^+, CN^+, C_2N^+, C_3N^+ - He collisions at $v_p = 2.2$ au

- The choice of projectiles extends previous studies conducted in the team on carbonaceous species such as C_n^+ (n=2-10) and $C_nH_m^+$ (n=1-4, m=1-4).

- We measured cross sections for basic single and multiple electron processes and the associated fragmentation. The goal is:

  - to test a simple collision model.
  - to test statistical fragmentation theories (in progress, not presented)
  - to valorise fragmentation data through construction of breakdown curves, to be used for instance in astrochemistry.



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Conclusions and Perspectives

- We measured cross sections of various electronic processes in N^+,CN^+,C_2N^+,C_3N^+ - He collisions at intermediate velocity. It compared well with predictions of the IAE/CTMC model with the exception of anionic production CS.

  We measured the associated fragmentation and used the dissociation Branching ratios in the context of astrochemisty.

In the future it will be interesting to :

- Understand the strong EXP/model disagreement for anionic production CS. A better calculation with non-independent electron on Helium for $CN^+ \rightarrow CN^-$ would be desirable.
- More generally a CTMC approach for molecule-atom collision on these systems would be of great interest.
- Compare measured dissociation BR with results of the M3C statistical fragmentation theory (Aguirre et al., 2017 and in progress)
- For astrochemical applications, choose to work with more strategic species. From the experimental point of view, many more species will be available with the development of a new source of cations at the Tandem accelerator.